

## Neva Krien

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## Related Projects

### Parallel Datalog Interpreter

[Source](#)

A standards-oriented Datalog compiler and query engine written in Rust. It uses magic sets, custom data structures, and Rayon for parallel evaluation.

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### Turing Compiler

[Source](#)

A C++11 and C99 project that compiles a Turing-machine-like language into x86 and ARM assembly, with loop inlining, constant-branch removal, and dead-code elimination.

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### SourceViewer

[Source](#)

A Rust TUI for exploring disassembly and source mappings from DWARF debug information, designed to work across languages and build systems.

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### Stack-Backed Memory Utilities

[Source](#)

Rust arenas and vector-like data structures for working with caller-provided memory without directly allocating on the heap.

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### GPU Cellular Automata

[Source](#)

A Rust and wgpu simulation engine that runs cellular automata entirely on the GPU using separate WGSL shaders for simulation and rendering.

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## Benchmarking Errors

[Source](#)

A collection of demonstrations of unusual facts about computer architecture and performance, such as the order of object files meaningfully impacting performance. Each example includes statistical analysis.

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## Fayen-lang

[Source](#)

A simple functional programming language with a stack-based VM written in Rust.

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## Dynamic Parser

[Source](#)

An LL(1) parser generator that supports macros and runtime grammar changes, including handling C's type grammar ambiguity by updating the grammar dynamically.

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## Open Source & Language Community

- Collaborate on API design and implementation for the Rust diagnostics library [Ariadne](#), across its earlier [GitHub repository](#) and current development on Codeberg.
  - Helped contributors across several beginner-led programming-language projects through guidance, debugging, and occasional code contributions; also worked on an optimization for an open-source [B compiler](#).
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## Intel Internship

### AI Deployment

Reference: Guy Tamir (Intel), 054-788-5085

Evaluated AI models, primarily LLMs, and explored deployment methods for Intel laptops under VRAM constraints. This work also included co-hosting an in-person workshop with Guy Tamir for 20 participants.

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## C++ Code Research

[One of the research group's papers](#)

I have been part of a research group (Intel + Technion collaboration) focusing on LLMs specializing in C++ code, particularly high-performance C++ such as MPI and OpenMP.